

plan: dominated convergence theorem

Fourier series, pointwise convergence theorem, version 1

Dominated convergence theorem

$$(-\infty \leq a < b \leq \infty)$$

if $f_n \rightarrow f$ pointwise and $\exists g \in L^1(a, b)$ s.t.

$$|f_n(x)| \leq g(x) \text{ for all } n \text{ and } x \in (a, b)$$

then $f(x)$ is integrable (i.e., $f \in L^1$) and

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \int_a^b f_n(x) dx$$

note: $g(x)$ is real-valued and non-negative

It's essential to have a dominator:

Example: $f_n(x) = \begin{cases} 1 & n \leq x < n+1 \\ 0 & \text{o.w.} \end{cases}$

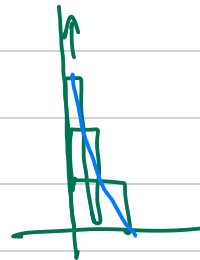


$$g(x) \geq 1 \text{ for } x \geq 0 \\ g \notin L^1(0, \infty)$$

$$f_n \rightarrow 0 \text{ pointwise but } \int f_n(x) dx = 1 \not\rightarrow \int f(x) dx = 0$$

Example:

$$f_n(x) = \begin{cases} n & 0 \leq x \leq \frac{1}{n} \\ 0 & \text{o.w.} \end{cases}$$



$$g(x) = n, \frac{1}{n+1} < x \leq \frac{1}{n} \\ > \frac{1}{x} - 1 \\ g \notin L^1(0, 1)$$

$$f_n \rightarrow 0 \text{ pointwise but } \int f_n(x) dx = 1 \not\rightarrow \int f dx = 0$$

Corollary: (book's version)

If $f_n \rightarrow f$ pointwise and $\exists g \in L^2$ s.t. $|f_n(x)| \leq g(x) \quad \forall n, x$
then $f_n \rightarrow f$ in L^2 .

proof: Fix $x \in (a, b)$. first we show $|f(x)| \leq g(x)$.

note that $||f_n(x)| - |f(x)|| \leq |f_n(x) - f(x)| \rightarrow 0$ as $n \rightarrow \infty$

↑ reverse triangle inequality (HW1 #1)

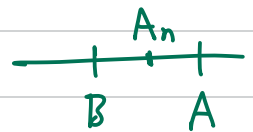
so $|f_n(x)| \rightarrow |f(x)|$ pointwise, and $|f_n(x)| \leq g(x) \quad \forall n$

In general, if $A = \lim_{n \rightarrow \infty} A_n$ exists and $A_n \leq B$

for all n , then $A \leq B$. Suppose to the contrary

that $A > B$. Let $\varepsilon = A - B$. Then $\exists N$ so that

for $n \geq N$, $|A_n - A| < \varepsilon$. But then



$$A - A_n \leq |A_n - A| < \varepsilon = A - B$$

so $-A_n < -B$ and $A_n > B$, contradicting $A_n \leq B$.

In our case, $A_n = |f_n(x)|$, $A = |f(x)|$, $B = g(x)$

and $|f(x)| \leq g(x)$ as claimed.

Since $h(x) = x^2$ is continuous at $x = 0$, $|f_n(x) - f(x)|^2 \rightarrow 0$ pointwise

$$|f_n(x) - f(x)|^2 \leq (|f_n(x)| + |f(x)|)^2 \leq (g(x) + g(x))^2 = 4g(x)^2$$

DCT gives $\int |f_n(x) - f(x)|^2 dx \rightarrow 0$ as $n \rightarrow \infty$.

$\therefore f_n \rightarrow f$ in L^2

in L^1 since $g \in L^2$
 $\int |g(x)|^2 dx < \infty$

Chapter 1. Fourier series.

$$V = L^2(-\pi, \pi), \quad \langle f, g \rangle = \int_{-\pi}^{\pi} f(x) \overline{g(x)} dx$$

$$V_N = \text{span} \{ e^{ikx} \}_{k=-N}^N \quad \dim V_N = 2N+1$$

claim: $\varphi_k(x) = \frac{1}{\sqrt{2\pi}} e^{ikx} \quad -N \leq k \leq N$

is an orthonormal basis for V_N .

proof: $\langle \varphi_n(x), \varphi_k(x) \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{inx} e^{-ikx} dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(n-k)x} dx = \textcircled{*}$

if $n=k$ then $\textcircled{*} = \frac{1}{2\pi} \int_{-\pi}^{\pi} 1 dx = 1$

otherwise $m = n-k \neq 0$ and

$$\textcircled{*} = \frac{1}{2\pi i m} e^{imx} \Big|_{-\pi}^{\pi} = \frac{e^{-im\pi}}{2\pi i m} (e^{2im\pi} - 1) = 0$$

$(e^{2\pi i})^m = 1$
↓

Orthogonal projection onto V_N : $P_N f(x) = \sum_{k=-N}^N \langle f, \varphi_k \rangle \varphi_k(x)$

$$P_N f(x) = \sum_{k=-N}^N c_k e^{ikx}, \quad c_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-ikx} dx$$

note: we absorbed $\frac{1}{\sqrt{2\pi}}$ from each φ_k into c_k

Next: as $N \rightarrow \infty$ and V_N gets bigger, determine in what sense

$$P_N f(x) \rightarrow f(x)$$

(pointwise? L^2 ? uniformly? how fast?)
Can we estimate the error $\|P_N f - f\|$?

Theorem: (pointwise convergence, version 1)

If $f(x)$ is continuous and 2π -periodic, then for each x_0 where f is differentiable, $P_N f(x_0) \rightarrow f(x_0)$.

proof: Let's write $S_N(x)$ instead of $P_N f(x)$. Then

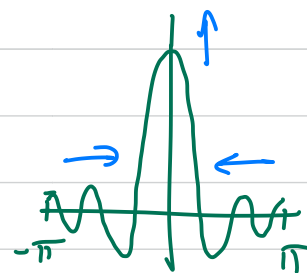
$$\begin{aligned} S_N(x_0) &= \sum_{k=-N}^N c_k e^{ikx_0} = \sum_{k=-N}^N \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-ikx} dx \right) e^{ikx_0} \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \sum_{k=-N}^N e^{ik(x_0-x)} dx \\ &= \int_{-\pi}^{\pi} f(x) D_N(x_0-x) dx \quad (\text{convolution}) \end{aligned}$$

Dirichlet kernel: $D_N(x) = \frac{1}{2\pi} \sum_{k=-N}^N e^{ikx}$

Intuition: As $N \rightarrow \infty$, $D_N(x)$ approaches a δ -function.

$$D_N(0) = \frac{1}{2\pi} \sum_{k=-N}^N e^0 = \frac{2N+1}{2\pi} \rightarrow \infty$$

$$\int_{-\pi}^{\pi} D_N(x) dx = \frac{1}{2\pi} \sum_{k=-N}^N \underbrace{\langle e^{ikx}, 1 \rangle}_{2\pi \delta_{k0}} = 1$$



spike gets taller and thinner as $N \rightarrow \infty$, integral remains 1.

Rigorous analysis:

$$2\pi D_N(x) = \sum_{k=-N}^N e^{ikx} = \sum_{j=0}^{2N} e^{i(j-N)x} = e^{-iNx} \sum_{j=0}^{2N} (e^{ix})^j$$

truncated geometric series:

$$(1 + z + z^2 + \dots + z^N)(1 - z) = \underbrace{1 - z}_0 + \underbrace{z - z^2}_0 - \dots - \underbrace{z^N + z^N}_0 - z^{N+1}$$

$$\text{so } \sum_{j=0}^N z^j = \begin{cases} N+1 & z=1 \\ \frac{1-z^{N+1}}{1-z} & z \neq 1 \end{cases}$$

I wrote this incorrectly on the board in class

$$2\pi D_N(x) = \underbrace{\left(\begin{matrix} -\pi \leq x \leq \pi \\ x \neq 0 \end{matrix} \right)}_{\substack{\text{ } \\ \text{ }}} e^{-iNx} \frac{1 - (e^{ix})^{2N+1}}{1 - e^{ix}} = \frac{e^{-iNx} - e^{i(N+1)x}}{1 - e^{ix}} \cdot \frac{e^{-ix/2}}{e^{-ix/2}}$$

$$= \frac{e^{-i(N+\frac{1}{2})x} - e^{i(N+\frac{1}{2})x}}{e^{-ix/2} - e^{ix/2}} = \frac{\sin((N+\frac{1}{2})x)}{\sin(\frac{x}{2})} \cdot \underbrace{\frac{-zi}{-zi}}$$

$$2\pi D_N(0) = 2N+1$$

$$\sin u = \frac{e^{iu} - e^{-iu}}{2i}$$

we cancelled these out

Note that $D_N(x)$ is even $\left(\begin{matrix} \text{ratio of odd functions} \\ \text{or use } 2\pi D_N(x) = 1 + 2 \sum_{k=1}^N \cos kx \end{matrix} \right)$

$$S_N(x_0) = \int_{-\pi}^{\pi} f(x) D_N(x_0 - x) dx$$

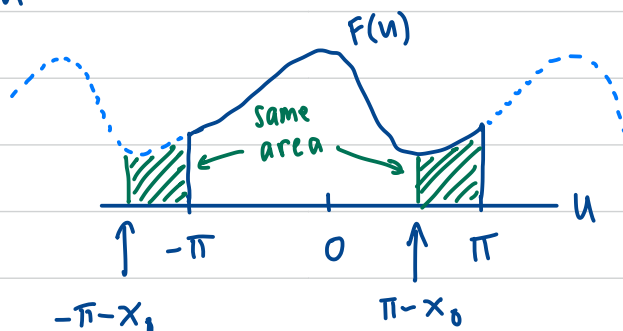
$$u = x - x_0 \\ du = dx$$

$D_N(x)$ is also 2π -periodic

$$= \int_{-\pi-x_0}^{-\pi+x_0} f(x_0+u) D_N(-u) du$$

$$= \int_{-\pi}^{\pi} f(x_0+u) D_N(u) du$$

$F(u) = f(x_0+u) D_N(u)$ is 2π -periodic. Can integrate over any period.



Finally, since $\int_{-\pi}^{\pi} D_N(u) du = 1$,

$$S_N(x_0) - f(x_0) = \int_{-\pi}^{\pi} \left[f(x_0 + u) - f(x_0) \right] \underbrace{D_N(u)}_{\frac{1}{2\pi} \frac{\sin((N+\frac{1}{2})x)}{\sin(x/2)}} du$$

we'll see next time that the

"bad denominator" $\sin \frac{x}{2}$ can be transferred to the difference $f(x_0 + u) - f(x_0)$ to obtain a continuous function $g(u)$ such that

$$S_N(x_0) - f(x_0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} g(u) \sin\left(N + \frac{1}{2}\right)u du$$

The Riemann-Lebesgue lemma now implies that

$$\lim_{N \rightarrow \infty} [S_N(x_0) - f(x_0)] = 0, \quad \text{i.e.} \quad \lim_{N \rightarrow \infty} S_N(x_0) = f(x_0)$$

version of R-L lemma we are using: if $g \in L^1(a, b)$ then

$$\lim_{k \rightarrow \infty} \int_a^b g(x) \sin kx dx = 0$$

where k is not restricted to be an integer.

(further details, discussion, and proof next time)